

CLASSIFICATION OF HAND-WRITTEN DIGITS

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1. INTRODUCTION

At first, we would like to give a brief introduction to the project.

2. LOGISTIC REGRESSION FOR MULTI-CLASS CLASSIFICATION

Logistic regression is a commonly used model for classification tasks. In our case we want to use logistic regression for multi-class classification, which is a generalization of the binary logistic regression.

In our case, we want to predict the class of a given 8×8 grayscale image of a hand written digit. One would expect that a given input can not be classified as more than one digit at the same time. Therefore, we want to compute the probability/score of a given input image to belong to every of the 10 classes (digits from 0 to 9) and assign the class with the highest probability/score to the input image.

2.1. Formalization of our problem.

An input image is represented as a vector $a_j \in \mathbb{R}^{64}$. Each entry of the vector is an integer between 0 and 16 and corresponds to the intensity of a pixel in the image with 0 being the lowest intensity and 16 being the highest intensity. Each image a_j in the dataset is associated with a label $y_j \in \{0, \dots, 9\}$, which is the digit that is represented in the image. In order to use the labels in our model, we encode them into one-hot vectors $p_j \in \{0, 1\}^{10}$, where the i -th entry is 1 if the label is i and 0 otherwise. [Dev]

We want to train a function $\varphi : \mathbb{R}^{64} \times (\mathbb{R}^{64} \times \mathbb{R})^{10} \rightarrow \mathbb{R}^{10}$ that takes an input image a_j and a parameter $x = (w_i, b_i)_{i=0}^9$ and outputs a vector of scores for each class. The score for class i is computed as $\varphi(a_j; x)_i = a_j^T w_i + b_i$. The $w_i \in \mathbb{R}^{64}$ is a weight vector and $b_i \in \mathbb{R}$ is a bias term for class i .

The training of the model consists of the following optimization problem

$$\underset{x \in (\mathbb{R}^{64} \times \mathbb{R})^{10}}{\text{minimize}} \sum_{j=1}^N L(p_j, \varphi(a_j; x)),$$

with $N \in \mathbb{N}$ being the number of training samples and L the cross entropy loss defined as

$$(2.1) \quad L(p, s) = \log \left(\sum_{l=0}^9 \exp(s_l) \right) - \sum_{i=0}^9 p_i s_i.$$

After the model has been trained, we can use it to classify new input images a by computing $\varphi(a; x)$ and assigning the class with the highest score to the input image, i.e.

$$\tilde{y} = \arg \max_{i \in \{0, \dots, 9\}} \varphi(a; x)_i.$$

The values of $\varphi(a_j; x)_i$ can be interpreted as probabilities of the input image a_j belonging to class i after applying the softmax function, i.e.

$$\sigma(s)_i = \frac{\exp(s_i)}{\sum_{l=0}^9 \exp(s_l)}.$$

2.2. Solvability.

The given optimization problem is a convex problem. This is because the log-sum-exp function is convex as we have already proved in the lecture. For reference one can also check [Boy09, pp. 72, 74]. Also the second term of our loss function (2.1) is linear and therefore convex. Also, φ is an affine mapping in x . And at the same time convex function over an affine function is also convex (cf. [Boy09, p. 67]).

In general however, the optimization problem is not necessarily solvable. This is because the loss function (2.1) is not strictly convex in its scores s and therefore not strictly convex in the parameter x .

This, is because the loss function (2.1) is invariant under the addition of the same constant to every score s_i . More precisely, for a constant $c \in \mathbb{R}$ we have

$$\log \left(\sum_{l=0}^9 \exp(s_l + c) \right) = \log \left(\exp(c) \sum_{l=0}^9 \exp(s_l) \right) = \log \left(\sum_{l=0}^9 \exp(s_l) \right) + c$$

and also

$$\sum_{i=0}^9 p_i(s_i + c) = \sum_{i=0}^9 p_i s_i + c.$$

Therefore, we have

$$L(p, s + c\mathbf{1}) = \log \left(\sum_{l=0}^9 \exp(s_l + c) \right) - \sum_{i=0}^9 p_i(s_i + c) = L(p, s) + c - c = L(p, s).$$

This implies that for a solution x of our optimization problem, we get arbitrary additional solutions, e.g. $x' = (w_i + c\mathbf{1}, b_i)_{i=0}^9$ for any constant $c \in \mathbb{R}$. Therefore, the optimization problem is not uniquely solvable.

REFERENCES

- [Boy09] Stephen P. Boyd. *Convex optimization*. Ed. by Lieven Vandenbergh. 7th ed. first published 2004. Cambridge University Press, 2009. 716 pp. ISBN: 9780521833783.
- [Dev] scikit-learn Developers. *Documentation of scikit-learn Machine Learning in Python*. Version 1.8.0 (stable). URL: <https://scikit-learn.org> (visited on 02/09/2026).